ELBO Gradient Calculation

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1. ELBO

In Normalizing Flows (NF), the optimization purpose is to maximize the Evidence Lower BOund (ELBO) with the available parameter θ

$$\mathrm{ELBO}(\theta) = \mathrm{E}q_{\theta(z)}[\log p(y|z) + \log p(z) - \log q_\theta(z)] \tag{1}$$

z: model parameters;

θ: Neural Network parameters;

y: observed data;

$L(\theta)$: ELBO, set as the objective function;

$p(y|z)$: likelihood function;

$p(z)$: model prior distribution;

$q_\theta(z)$: desired distribution to approximate the true posterior distribution in normalizing flows;

$\mathrm{E}q_{\theta(z)}$: average value;

As you can see that, the true model parameters are described by base distribution $q_\theta(z)$, while Neural Network parameters θ finally decide z. θ is the optimized during the iterations to find the maximum ELBO.

2. Normalizing Flows

The number ε is a random number that generated by probability density function $p_\varepsilon(\varepsilon)$. Usually, $p_\varepsilon(\varepsilon)$ can be set a simple Gaussian distribution (set as Base function)

$$p_\varepsilon(\varepsilon) = N(0, 1) \tag{2}$$

$f_\theta()$ is the normalizing flows that transform ε to z.

$$z = f_\theta(\varepsilon) \tag{3}$$

then

$$\mathrm{d}z = |\det(\frac{\partial f(\varepsilon)}{\partial \varepsilon})|\mathrm{d}\varepsilon \tag{4}$$

Since total probability must remain 1.0,

$$q_\theta(z)\,\mathrm{d}z = p_\varepsilon(\varepsilon)\,\mathrm{d}\varepsilon \tag{5}$$

Based on the above equations,

$$q_\theta(z) = p_\varepsilon(\varepsilon)\ |\det(\frac{\partial f(\varepsilon)}{\partial \varepsilon})|^{-1} \tag{6}$$

The above is called change of variables rule. And

$$\log q_\theta(z) = \log p_\varepsilon(\varepsilon) - \log |\det(\frac{\partial f(\varepsilon)}{\partial \varepsilon})| \tag{7}$$

3. Gradient calculation of ELBO

We usually choose the gradient-based method to optimize ELBO. The gradient of ELBO w.r.t. θ:

$$\nabla_\theta \mathrm{ELBO}(\theta) = \nabla_\theta E_{q_\theta(z)}[\log p(y|z) + \log p(z) - \log q_\theta(z)] \tag{8}$$

Since $z = f_\theta(\varepsilon)$ with $\varepsilon \sim N(0, I)$, the gradient can be set as

$$\nabla_\theta \mathrm{ELBO}(\theta) = E_\varepsilon[\nabla_\theta (\log p(y|\, f_\theta(\varepsilon)) + \log p(f_\theta(\varepsilon)) - \log q_\theta(f_\theta(\varepsilon)))] \tag{9}$$

Each term has the following:

Likelihood gradient:

$$\nabla_\theta \log p(y|\, f_\theta(\varepsilon)) = \nabla_z \log p(y|z) * \nabla_\theta f_\theta(\varepsilon) \tag{10}$$

Prior gradient:

$$\nabla_\theta \log p(f_\theta(\varepsilon)) = \nabla_z \log p(z) * \nabla_\theta f_\theta(\varepsilon) \tag{11}$$

NF gradient:

$$\nabla_\theta \log q_\theta(f_\theta(\varepsilon)) = -\nabla_\theta \log |\det(\frac{\partial f(\varepsilon)}{\partial \varepsilon})| \tag{12}$$

Putting the above three gradients into $\nabla_\theta \mathrm{ELBO}(\theta)$

$$\nabla_\theta \mathrm{ELBO}(\theta) = E_\varepsilon[(\nabla_z \log p(y|z) + \nabla_z \log p(z)) * \nabla_\theta f_\theta(\varepsilon) + \nabla_\theta \log |\det(\frac{\partial f(\varepsilon)}{\partial \varepsilon})|] \tag{13}$$

4. The simplified expression of the gradients

4.1 $\nabla_z \log p(y|z)$

The data noise is usually set as Gaussian with $N(0, \sigma_d)$

$$\log p(y|z) = -0.5/\sigma_d^2 \,\|\, d(z) - y \,\|^2 + \mathrm{const} \tag{14}$$

$d(z)$ is the forward calculated data. Then

$$\nabla_z \log p(y|z) = -1/\sigma_d^2 (\frac{\partial d(z)}{\partial z})^T (d(z) - y) \tag{15}$$

$\frac{\partial d(z)}{\partial z}$ is the low-efficient problem for the whole gradient calculation, which we choose to use adjoint-state method instead of explicitly calculating it.

4.2 $\nabla_z \log p(z)$

The prior model is usually set as Gaussian distribution $\log p(z) = N(\mu, \Sigma)$, we have

$$\log p(z) = -0.5 * (z-\mu)^{\mathrm{T}} \Sigma^{-1}(z-\mu) + \text{const}$$

$$\nabla_z \log p(z) = -\Sigma^{-1}(z-\mu) \tag{16}$$

So, the Gaussian distribution can make the gradient analytically calculated. If it's uniform distribution, the gradient $\nabla_z \log p(z)$ should be 0.0. Often, $\Sigma = \sigma_z^2 \mathrm{I}$, $\nabla_z \log p(z) = -1/\sigma_z^2(z-\mu)$.

4.3 $\nabla_z \log q_\theta(z)$

Based on formulas (3), ε can be expressed as

$$\varepsilon = f_\theta^{-1}(z) = g_\theta(z) \tag{17}$$

Based on change-of-variables rule,

$$\left(\frac{\partial f(\varepsilon)}{\partial \varepsilon}\right)^{-1} = \frac{\partial g(z)}{\partial z}$$

$$\left|\det\left(\frac{\partial f(\varepsilon)}{\partial \varepsilon}\right)\right|^{-1} = \left|\det\left(\frac{\partial g(z)}{\partial z}\right)\right| \tag{18}$$

Formula (7) can be written as

$$\log q_\theta(z) = \log p_\varepsilon(g_\theta(z)) + \log \left|\det\left(\frac{\partial g(z)}{\partial z}\right)\right| \tag{19}$$

So

$$\nabla_z \log q_\theta(z) = (\partial g_\theta(z)/\partial z)^{\mathrm{T}} * \nabla_\varepsilon \log p_\varepsilon(\varepsilon) + \nabla_z \log \left|\det\left(\frac{\partial g(z)}{\partial z}\right)\right| \tag{20}$$

As $p_\varepsilon(\varepsilon) = N(0, 1)$, $\nabla_\varepsilon \log p_\varepsilon(\varepsilon) = -\varepsilon$

$$\nabla_z \log q_\theta(z) = -(\partial g_\theta(z)/\partial z)^{\mathrm{T}} * \varepsilon + \nabla_z \log \left|\det\left(\frac{\partial g(z)}{\partial z}\right)\right| \tag{21}$$

5. Summary

Now, let's go back to see the ELBO gradient in formula (13) with the analytical expression in (15), (16) and (21), you will find that only $\nabla_\theta f_\theta(\varepsilon)$ and its determination $\det\left(\frac{\partial f(\varepsilon)}{\partial \varepsilon}\right)$ are unsolvable now, which they are both related to the Neural Network (NN) in Normalizing Flows (NF). That's why we should take care to choose the available NN/NF with accurate gradient/determination calculation. As you can see, Affine function, Coupling flows and Autoregressive flows have the advantage of accurately calculating gradient/determination and powerful expressive ability, so that's it.